# Coding & Solution

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| **Date** | 29 June 2026 |
| **Team ID** | XXXXXXXXXX |
| **Project Name** | Credit Card Approval Prediction |
| **Maximum Marks** | 5 Marks |

**Coding & Solution**
This section evaluates the quality and completeness of the implemented code and the overall solution. The submission demonstrates working functionality, clean coding practices, and alignment with the defined functional and non-functional requirements

**Instructions:**

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

**Solution Summary**

| Field | Details |
|---|---|
| Repository Link/URL | https://github.com/sowmyaperike/Credit-Card-Approval-Prediction |
| Programming language(s) | Python, HTML,CSS, JavaScript |
| Framework(s) Used | Flask, Scikit-learn, Pandas |
| Key Features Implemented | Credit card approval prediction, data preprocessing, feature engineering, one-hot encoding, feature alignment, Random Forest classification, risk audit report generation, prediction logging using JSON, responsive web interface, Docker support |
| Pending / Incomplete Features | User authentication, database integration, cloud deployment, email notification, model retraining with live data |
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To run this project, first clone the repository using "git clone https://github.com/sowmyaperike/Credit-Card-Approval-Prediction Setup / Run Instructions running "pip install -r requirements.txt" in the terminal. Once the installation is complete, make sure you are in the project directory and start the application by executing "python main.py".

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**Code Quality Checklist**

| S.No | Criteria | Status (Yes / No) |
|---|---|---|
| 1 | Code is modular and organized into functions / classes | Yes |
| 2 | Meaningful variable and function names are used | Yes |
| 3 | Code includes comments / documentation where necessary | Yes |
| 4 | Error handling is implemented for critical operations | Yes |
| 5 | The application runs without critical errors | Yes |
| 6 | Code is committed to a version control repository | Yes |

**Additional Notes / Comments:**

- The application follows a modular architecture, separating preprocessing, prediction, logging, and frontend components.
- A **Random Forest** machine learning model is used to provide accurate and reliable credit card approval predictions.
- Input data is preprocessed and aligned with the trained model before prediction to ensure consistency.
- Prediction results are accompanied by a **Risk Audit Report**, improving transparency and helping users understand the decision.

- The project is designed for future enhancements such as authentication, cloud deployment, and database integration while maintaining clean and reusable code.